

Web Technology



Introduction to React

Rourab Paul, Subin Sahayam, Assistant Professor,

**Department of Computer Science and Engineering
Shiv Nadar University**

React

- **React – JavaScript Library**
 - **Front-end Development**

React

- **React – JavaScript Library**
 - **Front-end Development**
- **Facebook Software Engineer - Jordan Walke**
- **React – Not a Framework like Angular**
- **React.js or ReactJS**
- **Build and Manage Efficient UI**



React

- **HTML creates the structure.**
- **CSS creates the design.**
- **JavaScript adds behavior.**
- **React helps manage large, dynamic user interfaces.**

Without React

```
<body>
  <div class="container">
    <h2>Without React (Vanilla JavaScript)</h2>
    <h1 id="count">0</h1>
    <button onclick="increase()">Increment</button>
  </div>
<script>
  let count = 0;
  function increase() {
    count++;
    // Manual DOM update
    1)Find the element with id='count'
    const element = document.getElementById("count");
    2)Replace its HTML content with 1
    element.innerHTML = count;
    3) Browser repaints that part of the page
    console.log("Count =", count);
  }
</script>
</body>
```

[Sample Code](#)

With React

```
<body>
  <div id="root"></div>
  <script type="text/babel">

    function App() {
      const [count, setCount] = React.useState(0);
      return (
        <div className="container">
          <h2>With React</h2>
          <h1>{count}</h1>
          <button onClick={() => setCount(count + 1)}>
            Increment
          </button>
        </div>
      );
    }

    const root = ReactDOM.createRoot(document.getElementById("root"));
    root.render(<App />);
  </script>
</body>
```

Virtual DOM

```
<div className="container"> // JSX
  <h2>With React</h2>       // JSX
  <h1>{count}</h1>           // JSX with
                             JavaScript expression
  <button onClick={...}>     // JSX with event
    Increment               handler
  </button>
</div>
```

JSX is used to create Virtual DOM objects.

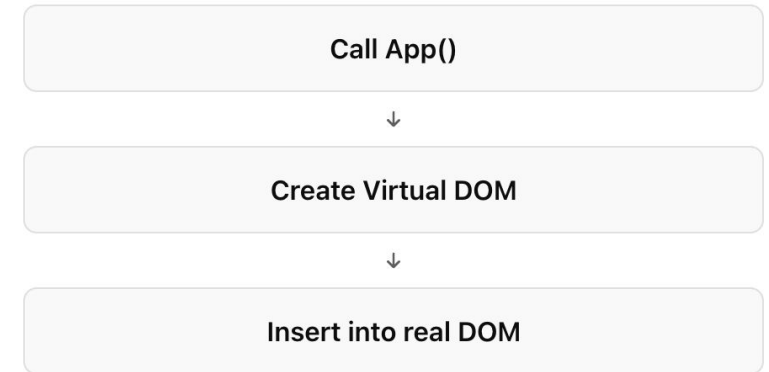
Create React root
React now calls App()

With React

```
<body>
  <div id="root"></div>
  <script type="text/babel">
    function App() {
      // State
      const [count, setCount] = React.useState(0);
      return (
        <div className="container">
          <h2>With React</h2>
          <h1>{count}</h1>
          <button onClick={() => setCount(count + 1)}>
            Increment
          </button>
        </div>
      );
    }
    const root = ReactDOM.createRoot(document.getElementById("root"));
    root.render(<App />);
  </script>
</body>
```

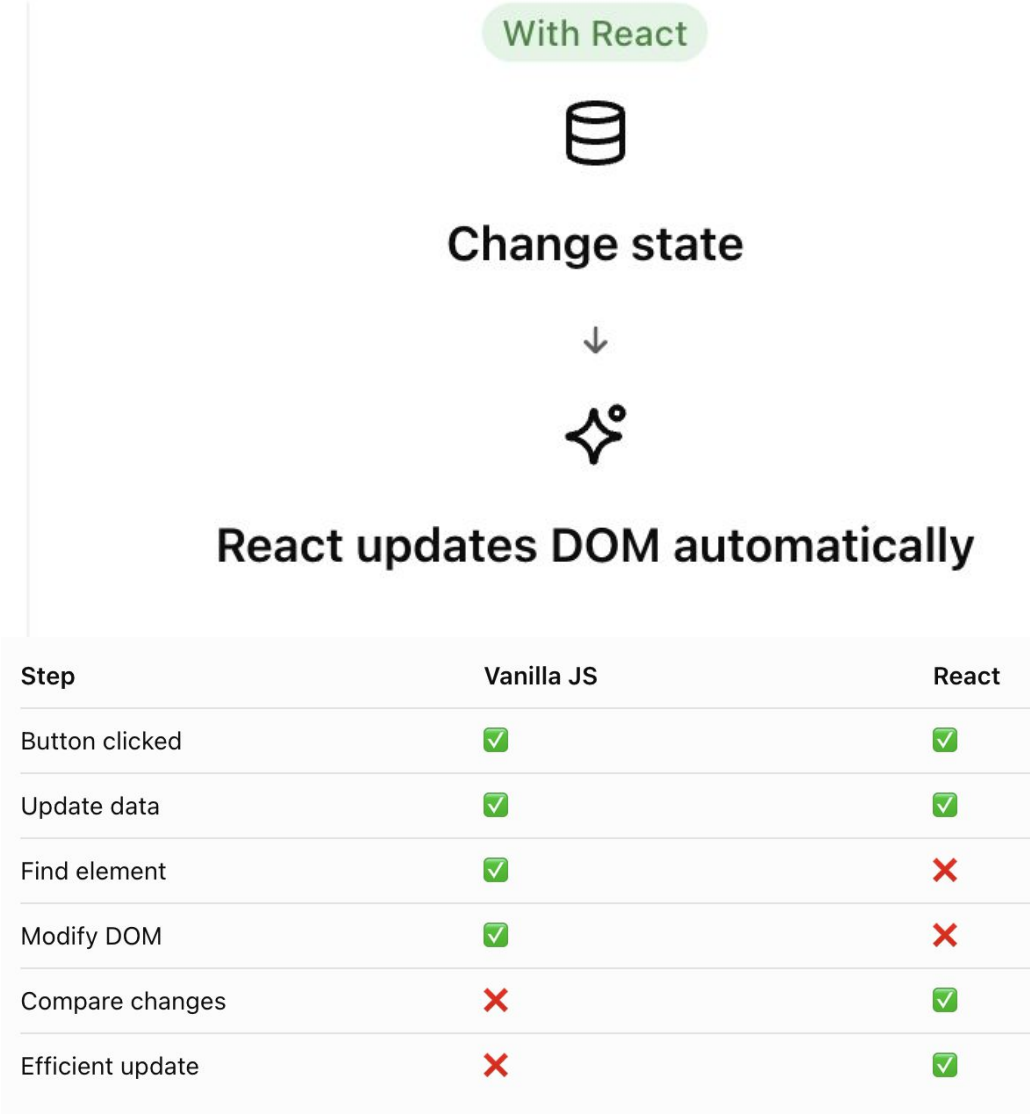
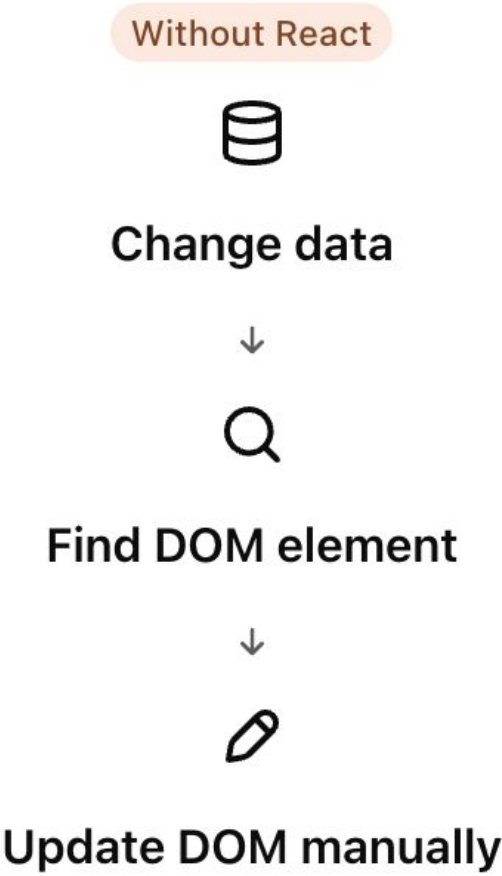
Virtual DOM

Create React root
React now calls App ()

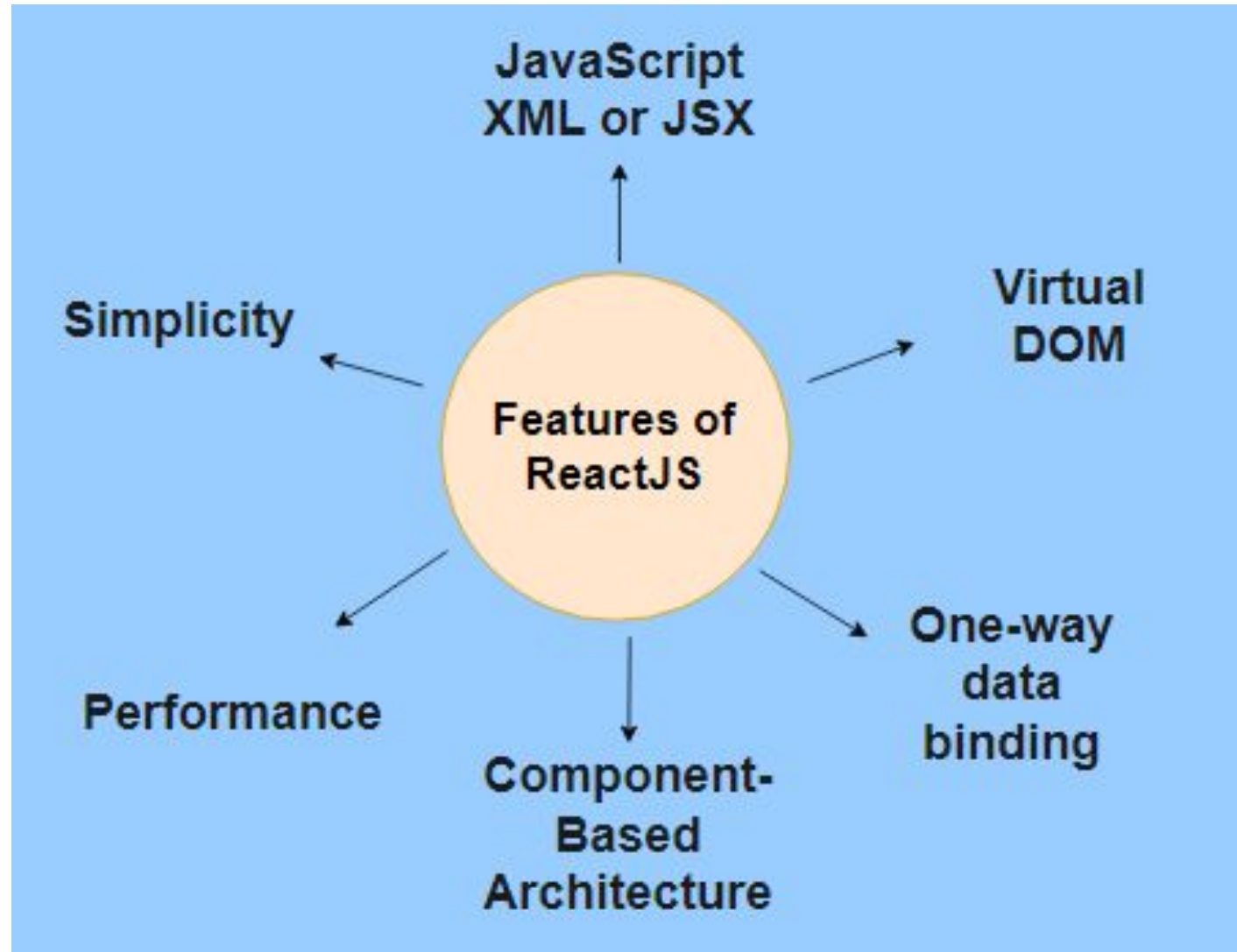


[Sample Code](#)

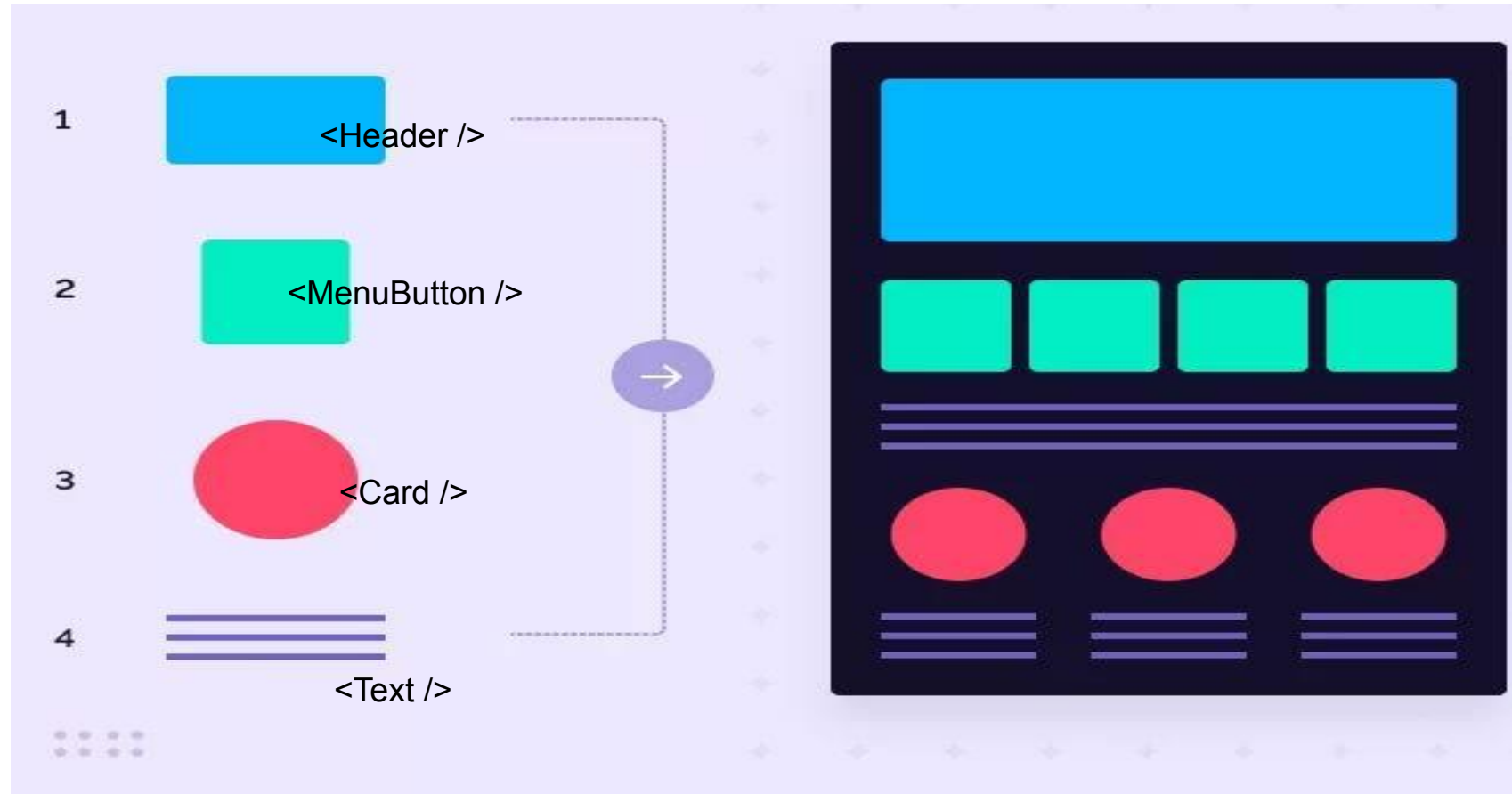
With & Without React



React Features



React Features - Components



with react

```
function Header() { return
<header>My Site</header>; }
```

```
function Card() { return <div
className="card">Product</div>
```

```
function App() { return ( <>
<Header />
<Card />
<Card />
<Card />
</> ); }
```

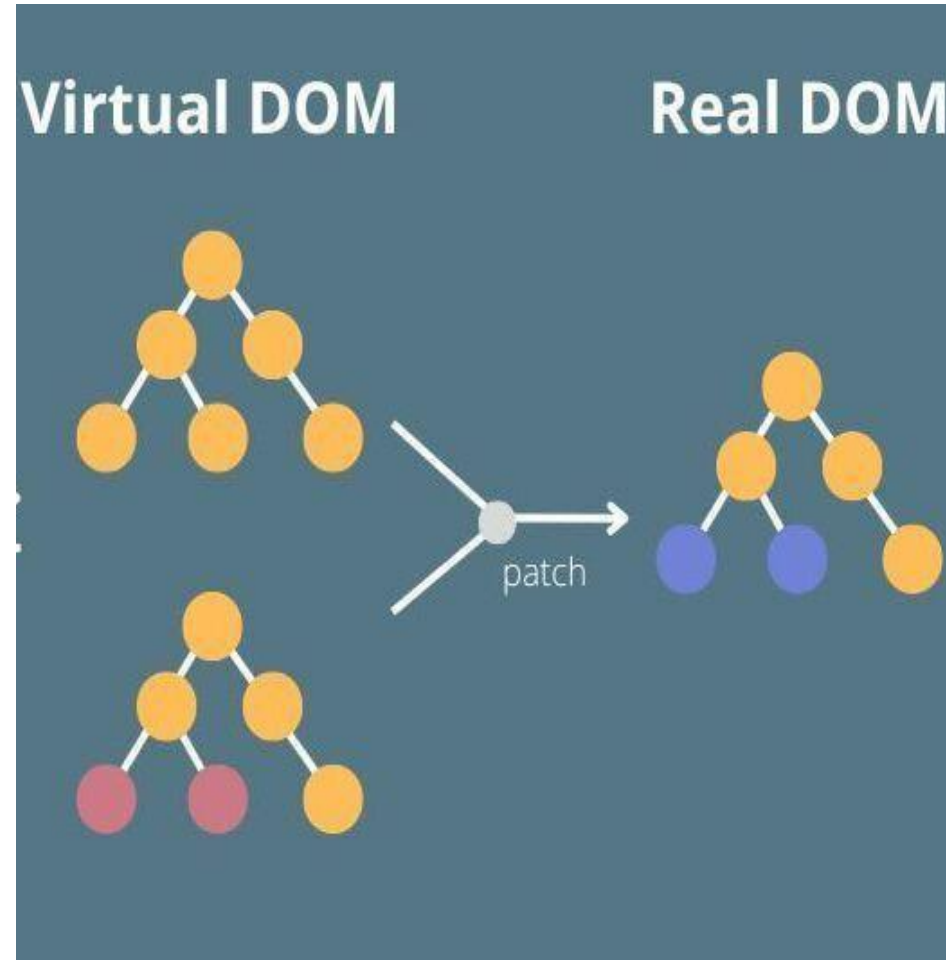
without react

```
<header>...</header>
<nav>...</nav> <div
class="card">...</div> <div
class="card">...</div> <div
class="card">...</div>
```

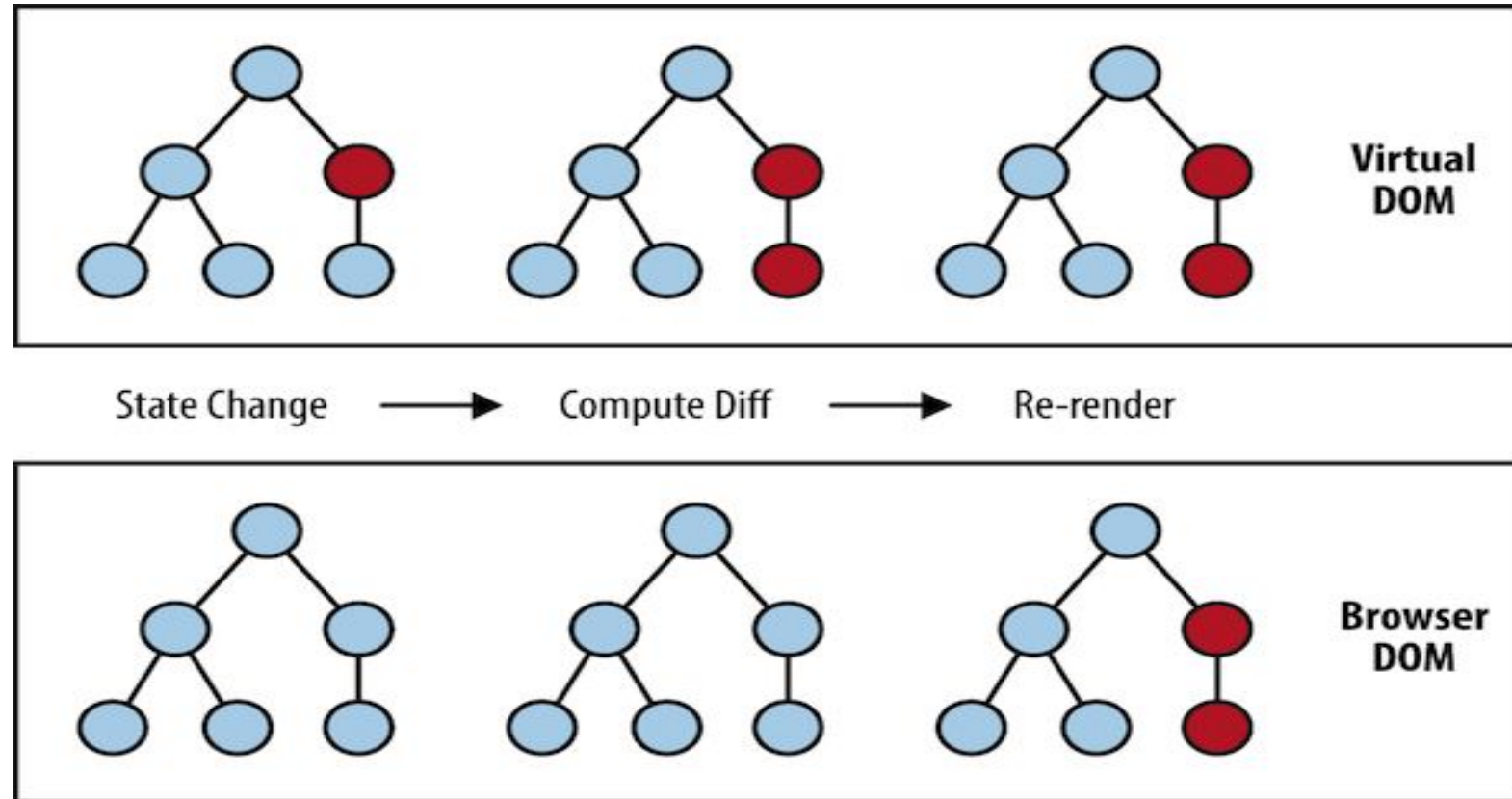
React Features - JSX

```
JS firstcomponent.js ✕  
▸ firstreactproject ▸ src ▸ components ▸ JS firstcomponent.js ▸  
1  
2 function FirstComponent() {  
3   return (  
4     <h1> My First Component</h1>   
5   );  
6 }  
7  
8 export default FirstComponent;
```

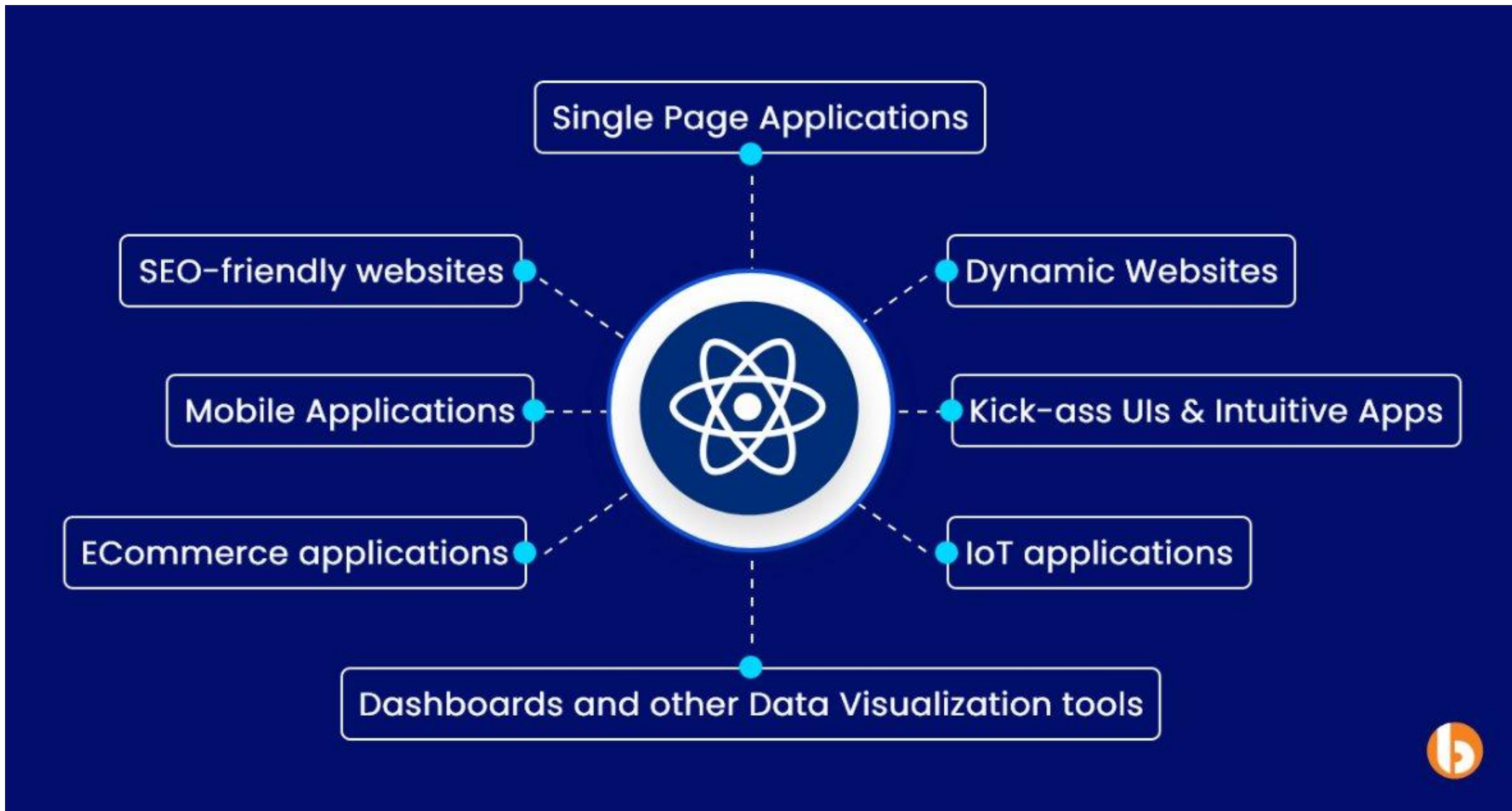
React Features - Virtual DOM



React Features - Virtual DOM



React Applications



1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**

THANK YOU

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